IBUPROFEN

Avoid in 3rd trimester

Take with food

CHEMICAL NAME (RS)-2-(4-(2-methylpropyl)phenyl)propanoic acid
CLASS Non-selective NSAID (propionic acid derivative)
TRADE NAMES Brufen · Advil · Nurofen · Motrin

THERAPEUTIC USES / INDICATIONS

- Mild–moderate pain: dental, musculoskeletal, dysmenorrhoea, headache.
- Fever in adults and children.
- Inflammatory conditions: rheumatoid arthritis, osteoarthritis, soft-tissue injury.
- IV form: closure of patent ductus arteriosus in neonates.

ADMINISTRATION

- **PO:** tablets, suspension; give with or after food or milk.
- **Topical** gel for local muscle pain.
- **IV** infusion in hospital settings.

DOSAGES

- **Adult pain/fever:** 200–400 mg every 4–6 h; OTC max 1.2 g/day.
- **Adult inflammatory disease:** up to 2.4 g/day in divided doses (max 3.2 g/day under supervision).
- **Children (>3 months):** 5–10 mg/kg every 6–8 h; max 30–40 mg/kg/day.
- Use lowest effective dose for the shortest duration.

PHARMACOKINETICS

- **Onset:** 30–60 min. **Peak:** 1–2 h. **Duration:** 4–6 h (anti-inflammatory effect needs regular dosing for days).
- Highly protein-bound (99%); metabolised in the liver (CYP2C9).
- Excreted in urine. **t½:** about 2 h.

PHARMACODYNAMICS

- Reversibly inhibits **COX-1 and COX-2** → ↓ prostaglandin synthesis.
- Effects: analgesic, antipyretic and anti-inflammatory.
- COX-1 inhibition reduces gastric protection and platelet aggregation.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Dyspepsia, nausea, abdominal pain, heartburn.
- Headache, dizziness.
- Fluid retention, mild BP rise.

ADVERSE EFFECTS (SERIOUS)

- GI ulceration and bleeding.
- Acute kidney injury (especially with dehydration).
- Bronchospasm in aspirin-sensitive asthma.
- ↑ risk of MI and stroke at high doses; hyperkalaemia.

CONTRAINDICATIONS & CAUTIONS

- Active peptic ulcer or GI bleeding; history of NSAID-induced bleeding.
- Severe heart, renal or hepatic failure.
- Asthma or allergy triggered by aspirin/NSAIDs.
- Third trimester of pregnancy (premature closure of ductus arteriosus).
- **Caution:** elderly, dehydration, hypertension.

KEY INTERACTIONS

- **Anticoagulants, SSRIs, steroids:** ↑ bleeding risk.
- **ACE inhibitors/ARBs + diuretics:** ↑ renal failure risk ("triple whammy").
- **Lithium, methotrexate:** ↑ levels.
- **Aspirin:** may reduce its antiplatelet effect.

NURSING MANAGEMENT

- Assess pain/temperature before and 1 h after the dose.
- Ask about ulcers, asthma, kidney disease and anticoagulant use.
- Give with food; consider PPI cover in high-risk patients.
- Monitor for black stools, haematemesis, oedema and urine output.
- Long-term use: monitor BP, renal function and CBC.

PATIENT EDUCATION

- Take with food or milk; do not lie down right after.
- Report black stools, vomiting blood or swelling.
- Avoid combining with other NSAIDs (including aspirin) or alcohol.
- Drink enough fluids when febrile or unwell.

KEY POINTS

- Analgesic, antipyretic **and** anti-inflammatory.
- Avoid in 3rd trimester and NSAID-sensitive asthma.
- Major risks: GI bleeding, kidney injury.
- Child dose 5–10 mg/kg every 6–8 h.

SOURCES CHECKED **[4]** Ibuprofen 200 mg solution for infusion (UK SmPC) **[5]** Ibuprofen 100 mg/5 ml Oral Suspension (UK SmPC) **[6]** Ibuprofen, oral (US label) **[7]** Caldolor, IV (US label)
Full references p. 432–457